

The Political Economy of Local Land-Use Regulation

The theory line: fragmented regulation as a fiscal-federalism breakdown

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August 2026

1 Purpose

The project studies the *market for housing regulation*. Its empirical spine is an item-level dataset of discretionary land-use decisions, built by parsing SF Planning Commission minutes and being extended to the roughly 109 land-use regulators of the nine-county Bay Area, paired with the by-right zoning envelope and with dated state-preemption exposure.

This directory is the **theory line**, not a data probe. It holds the model that tells the empirical work what to measure and why: it defines the two regulatory margins the dataset is built to separate, states the propositions the estimation is meant to test, and names the counterfactuals the measurement is ultimately for. Its claim on the project is that a purely descriptive account of land-use stringency cannot answer either question the project cares about — how large the fragmentation distortion is, and how much of state preemption localities claw back — because both are counterfactual, and both require an equilibrium object rather than a level.

2 What is in this directory

3 The model

3.1 Environment

One metro (the Bay Area) contains J jurisdictions $j = 1, \dots, J$, grouped into $S \ll J$ housing submarkets $s(j)$, all drawing on a *single* regional labor market. A worker employed anywhere in the metro earns the regional wage $w_m = w(N)$ with $w' > 0$ and $N \equiv \sum_j N_j$, which carries the agglomeration rent. This wage coupling is what makes the model's central asymmetry structural rather than assumed: a household's income depends on the *metro*, not on the jurisdiction it sleeps in.

A household of type θ living in j has preferences

$$U = c + \alpha \ln h - \gamma g(D_j), \quad g' > 0, \quad y(\theta) = w_m \theta,$$

where $D_j = N_j/\bar{L}_j$ is local density and $-\gamma g(D_j)$ is the *local* disamenity of density. Because low density enters as a normal good, high- θ residents are the natural demanders of restriction. Within submarket s , market clearing gives the price index

$$p_s = \frac{\alpha N_s}{H_s}, \quad N_s = \sum_{j: s(j)=s} N_j, \quad H_s = \sum_{j: s(j)=s} H_j. \quad (1)$$

File	What it is	How it was produced
<code>toy_model.tex</code>	The model note, “Fragmented Housing Regulation as a Fiscal-Federalism Breakdown” (June 2026): environment, two margins, homevoter objective, three propositions, identification, data mapping.	Written by hand; the analytical core of the project. Compiles against <code>toy_model.bib</code> with <code>natbib/plainnat</code> .
<code>toy_model.pdf</code>	Compiled note.	<code>pdflatex</code> + <code>bibtex</code> .
<code>toy_model.bib</code>	Bibliography (Oates, Besley–Coate, Fischel, Glaeser, Hsieh–Moretti, Bartik et al., Brueckner).	Hand-assembled.
<code>guren_meeting_slides.tex</code>	A 10-frame Beamer deck presenting the same framework for the Adam Guren meeting: the two margins, the three-layer structure (demand / supply / regulation), the reaction function, and four explicit questions for the advisor.	Written 2026-06-16 alongside the demand-data report.
<code>guren_meeting_slides.pdf</code>	Compiled deck.	<code>pdflatex</code> .
<code>.nav</code> , <code>.snm</code>	Beamer navigation auxiliaries.	Compilation by-products.

Equation (1) is the externality. Housing built in any jurisdiction of s lowers the price faced by incumbents in *every* jurisdiction of s , while density is local. The cost of approval is borne locally; the benefit — lower submarket prices, more regional employment and hence a higher w_m — is shared. That is exactly the spillover condition the Decentralization Theorem excludes, so decentralized provision of regulation is inefficient.

3.2 Two margins of regulation

The model’s key structural move comes from an empirical fact visible in the minutes: among projects that reach a vote, approval is nearly universal. Restriction does not operate through denial at the hearing. It operates before and around it, on two margins.

- **The by-right envelope $\bar{z}_j$.** The zoning map’s use and height/bulk districts define what a project may obtain *ministerially* — with certainty, no hearing. Projects inside the envelope never appear on the commission calendar. This is the margin that LLM-based zoning measurement captures, and the margin state preemption targets.
- **The discretionary friction τ_j .** A project exceeding the envelope enters discretionary review and is built only after paying a *discretionary tax* whose components the minutes record: process exposure (the share of activity forced off the by-right path), mobilized opposition (the operative form of $g(D_j)$, measured by stance-marked speaker counts), conditioning (approval-

with-conditions), delay (continuances), and rare denial. This is the margin LLM zoning measurement *cannot* see and preemption leaves untouched.

Effective stringency is then $r_j = g(\bar{z}_j, \tau_j)$, increasing in the tightness of the envelope (a smaller $\bar{z}_j$ pushes more activity into discretion) and in the discretionary tax. The two margins are *substitutes* in producing restriction but face different costs — and preemption acts on one and not the other. This definition is what makes leakage a coherent object; it is the piece the descriptive literature lacks.

3.3 The homevoter’s problem

Regulation is chosen by a vote-maximizing local authority answering to incumbent homeowners (share λ_j , political weight ϕ_j). The two margins map to two bodies — the council sets the slow map $\bar{z}_j$, the planning commission sets the fast discretionary tax τ_j — which resolves the usual unitary-authority shortcut. Taking neighbors’ policies as given, an incumbent’s payoff is

$$W_j^{\text{own}} = \underbrace{p_{s(j)}(\bar{z}_{-j}, \tau_{-j}; \bar{z}_j, \tau_j)}_{\text{submarket asset value}} - \underbrace{\gamma g(D_j(\bar{z}_j, \tau_j))}_{\text{local disamenity}}, \quad (2)$$

where $-j$ is the profile of all other jurisdictions and only co-submarket neighbors ($s(k) = s(j)$) enter with nonzero weight. The authority maximizes

$$\Omega_j = \phi_j \lambda_j W_j^{\text{own}} + (1 - \phi_j \lambda_j) W_j^{\text{ent}} \quad \text{over } (\bar{z}_j, \tau_j), \quad (3)$$

so $\phi_j \lambda_j$ is the model’s homevoter weight and W^{ent} the payoff of entrants.

3.4 Solution concept

Because $p_{s(j)}$ in (2) depends on *neighbors’* policies, (3) is not a decision problem but a game. The solution is the Nash equilibrium in policies, characterized by a reaction function on each margin,

$$(\bar{z}_j, \tau_j) = R_j(\bar{z}_{-j}, \tau_{-j}), \quad -j = \text{co-submarket neighbors only},$$

and it is compared against the allocation chosen by an authority that internalizes the submarket (or region) — the planner who values entrants’ access to w_m and the cross-jurisdiction price externality in (1).

4 Preliminary results

4.1 The free-riding wedge

The mechanism is an attenuation result. The effect of j ’s own approvals on the price its homeowners care about is diluted by submarket size:

$$\frac{\partial p_{s(j)}}{\partial \text{supply}_j} = -\frac{p_s}{H_s}. \quad (4)$$

A jurisdiction that is small within its submarket bears the *full* local disamenity of any approval but captures only a $1/H_s$ share of the price relief; the rest leaks to co-submarket neighbors. It therefore under-supplies and free-rides on their construction. Equation (4) is the microfoundation of the fragmentation externality and drives everything below.

Proposition 1 (Fragmentation over-restricts; the distortion is the Oates wedge). Holding $(\phi_j \lambda_j, \gamma, \bar{L}_j)$ fixed, a jurisdiction’s privately optimal restriction is *increasing in the number of jurisdictions sharing its submarket*. As j becomes atomistic the price-relief motive vanishes, the local-disamenity motive dominates, $\bar{z}_j$ tightens and τ_j rises. The *distortion* is the gap between decentralized Nash regulation $\{(\bar{z}_j^*, \tau_j^*)\}$ and the regulation an internalizing authority would choose. This is the model’s answer to Question 1, and it is precisely the inefficiency the Decentralization Theorem’s spillover condition rules in.

Proposition 2 (Regulation is strategically substitutable within a submarket). For co-submarket jurisdictions, if a neighbor loosens (raises $\bar{z}_k$ or lowers τ_k), the submarket price falls, which *raises* a homevoter-pivotal j ’s incentive to restrict: j tightens. Neighbors’ regulations are strategic substitutes, each free-riding on the others’ supply, and the Nash equilibrium delivers less regional construction than the internalized optimum.

Why Proposition 2 is the centerpiece. The reaction-function slope is not merely an empirical regularity; it is the direct test of the federalism breakdown. Under the Decentralization Theorem’s efficient case — no spillovers — jurisdictions would not respond to one another’s regulation at all, and the cross-jurisdiction slope would be zero. A robustly *nonzero* slope is the spillover made visible: its sign and magnitude establish that the Oates condition fails and quantify how badly. This is also why the project must scale beyond San Francisco: single-jurisdiction data cannot, even in principle, reveal a reaction function.

Proposition 3 (Leakage: substitution from envelope to discretion). State preemption — ministerial-approval mandates, by-right upzoning, the builder’s remedy — raises the marginal cost of a tight *envelope* while leaving the cost of the *discretionary* tax unchanged. With $r_j = g(\bar{z}_j, \tau_j)$ and the margins substitutable, a preemption that raises $\bar{z}_j$ (the map loosens, as intended) induces a homevoter-pivotal jurisdiction that still demands restriction to raise τ_j in compensation: more items conditioned, more opposition entertained, more continuances. Effective restriction falls by *less* than the change in $\bar{z}_j$ alone implies, and — the sharp testable implication — **the conditional approval rate need not move at all**. Leakage is the share of intended loosening offset on the untouched margin, running from 0 (preemption fully effective) to 1 (fully clawed back). This is the answer to Question 2.

Proposition 3 reframes preemption as a federalism question rather than a mechanical policy: partial centralization fails not because centralization is wrong but because it is *incomplete*, leaving a decentralized margin through which the spillover reasserts itself. Proposition 2 compounds it — as preempted jurisdictions claw back via τ , co-submarket neighbors face a smaller price effect and adjust in turn.

4.2 Identification and the mapping to data

Both headline questions are counterfactual, and only *equilibrium* regulation is observed, never the objective that generated it; recovering $(\phi_j \lambda_j, g(\cdot), \text{demand} + w(N))$ is what licenses the counterfactuals. The reaction-function slope is consistent with rivals — yardstick competition, common shocks — so the note routes identification through the **preemption ramp**: the staggered, rule-driven timing of state mandates (builder’s-remedy exposure following Housing-Element non-compliance) is a supply-side shock plausibly orthogonal to a locality’s contemporaneous demand shock, moving $\bar{z}_j$ for reasons other than local taste.

Each primitive has an address in the data: $\bar{z}_j$ is read from zoning (largely available from existing LLM-based measurement); τ_j is built fresh from planning-commission minutes item by item; the preemption ramp is dated from HCD Housing-Element compliance records (the panel in `output/bay_area_recon/hcd_preemption_panel/`); prices and quantities come from transaction data. Proposition 2 becomes a spatial-lag regression of a locality’s discretionary tax on its co-submarket neighbors’, instrumented with the preemption variation. Proposition 3 becomes a difference-in-differences around preemption exposure with the *envelope* and the *discretionary tax* as *separate* outcomes — predicting the map loosens, discretion tightens, and the approval rate stays flat. A short panel straddling the 2017–2023 ramp suffices for both.

5 Status and open questions

1. **The model is written; nothing is estimated.** `toy_model.tex` is a complete analytical note with three stated propositions, but the propositions are argued rather than proved from an explicit solved equilibrium, and no parameter has been taken to data. All numbers in the project so far are data-feasibility counts, not estimates.
2. **The multi-jurisdiction extension is the binding gap.** The reaction function of Proposition 2 requires a panel of jurisdictions, and only SF minutes have been processed. The progress log carries this explicitly as the open build item: construct the multi-jurisdiction τ_j panel and integrate $\bar{z}_j$ for the strategic-interaction layer. Until then Proposition 2 is untestable and Proposition 1’s counterfactual has no estimated primitives.
3. **The by-right versus ministerial definition is unresolved** — and it is the model’s hinge. Whether a use is principally permitted (built with certainty, no hearing) as against conditionally permitted or subject to design review determines what $\bar{z}_j$ even means, and hence what the leakage test measures. This is a modeling decision, not a measurement; no further data work resolves it.
4. **Identification against the rivals is the acknowledged soft spot.** Separating the spillover from yardstick competition and common shocks is the question the Guren deck puts to the advisor directly, along with whether the leakage result (Q2) should stand alone as a first paper with the structural reaction function (Q1) as the extension.
5. **The dynamic ratchet is deliberately deferred.** A state-dependent version — regulation as path-dependent, with persistent or hysteretic preemption effects propagating across neighbors — is noted as a separate extension. It would need a long panel and confronts state dependence versus unobserved heterogeneity. Neither headline question depends on it, which is why the deep-history data workstream is scoped out.